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Aim:

Write a C program to find the Minimum Spanning Tree (MST) of a connected, weighted, undirected graph using Prim's Algorithm.

You are given a partially implemented driver code. Your task is to complete the missing logic in the **minKey(), prim()** functions. The program represents the graph using an adjacency matrix.

Functions to be Completed

1. minKey(): This function selects the vertex that has the minimum key value from the set of vertices not yet included in the MST.

```
int minKey(int key[], bool mstSet[], int vertices)
```

- key[] stores the minimum weight edge connecting a vertex to the MST
- mstSet[] keeps track of vertices already included in the MST
- The function returns the index of the vertex with the smallest key value that is not yet in the MST

2. prim(): This function implements Prim's Algorithm to construct the Minimum Spanning Tree.

```
void prim(int graph[V][V], int vertices)
```

- This function finds the MST of a graph by starting from one vertex and repeatedly adding the nearest (lowest weight) vertex until all vertices are connected.

Input Format:

- The program first prompts the user to enter an integer vertices represents the number of vertices in the following format:

```
No of vertices:
```

- Next line, the program prompts:

```
Adjacency matrix elements (row wise):
```

- The next vertices lines each contain vertices space-separated integers representing the adjacency matrix.
- The value at row i and column j denotes the weight of the edge between vertex i and vertex j.
- A value of 0 indicates that there is no edge between the corresponding vertices.

Output Format:

- After constructing the Minimum Spanning Tree, the program prints the header:

```
Edge  Weight
```

- Then, for each edge selected in the MST, print:

```
<u> - <v>  <weight>
```

Where:

- <u> is the parent vertex of v in the MST
- <v> is the current vertex

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- <weight> is the weight of the edge between vertices u and v

The edges are printed starting from vertex 1 up to vertices - 1, in the order they are added to the MST.

Constraints:

- $1 \leq \text{vertices} \leq 100$
- Edge weights are non-negative integers
- The graph is connected
- The adjacency matrix is symmetric (undirected graph)

Note:

- The MST always starts from vertex 0
- Refer to the visible test cases for better understanding.

Source Code:

minCostFinding.c

```
#include <stdio.h>
#include <stdbool.h>
#include <limits.h>
#define V 100

int minKey(int key[], bool mstSet[], int vertices) {
   // Write your code here...
   int min = INT_MAX, min_index;

   for (int v = 0; v < vertices; v++)
      if (mstSet[v] == false && key[v] < min)
         min = key[v], min_index = v;

   return min_index;



}
void printTree(int parent[], int graph[V][V], int vertices) {
    printf("Edge \tWeight\n");
    for (int i = 1; i < vertices; i++)
        printf("%d - %d \t%d \n", parent[i], i, graph[i][parent[i]]);
}

void prim(int graph[V][V], int vertices) {
   // Write your code here...

   int parent[V];
    int key[V];
    bool mstSet[V];

    for (int i = 0; i < vertices; i++)
        key[i] = INT_MAX, mstSet[i] = false;

    key[0] = 0;
    parent[0] = -1;
```


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```
    for (int count = 0; count < vertices - 1; count++) {
        int u = minKey(key, mstSet, vertices);
        mstSet[u] = true;

        for (int v = 0; v < vertices; v++)
            if (graph[u][v] && mstSet[v] == false && graph[u][v] < key[v])
                parent[v] = u, key[v] = graph[u][v];
    }

   printTree(parent, graph, vertices);

}

int main() {
    int vertices;
    int graph[V][V];

    printf("No of vertices: ");
    scanf("%d", &vertices);

    printf("Adjacency matrix elements (row wise):\n");
    for (int i = 0; i < vertices; i++) {
        for (int j = 0; j < vertices; j++) {
            scanf("%d", &graph[i][j]);
        }
    }

    prim(graph, vertices);

    return 0;
}
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
No of vertices: 5
Adjacency matrix elements (row wise): 0 0 4 0 0
0 0 5 3 0
4 5 0 0 0
0 3 0 0 2
0 0 0 2 0
Edge Weight
2 - 1 5
0 - 2 4
1 - 3 3
3 - 4 2